

R Notebook

```
library(phyloseq)
library(microbiome)
```

```
## Loading required package: ggplot2
```

```
##
## microbiome R package (microbiome.github.com)
##
##
## Copyright (C) 2011-2022 Leo Lahti,
## Sudarshan Shetty et al. <microbiome.github.io>
```

```
##
## Attaching package: 'microbiome'
```

```
## The following object is masked from 'package:ggplot2':
##
## alpha
```

```
## The following object is masked from 'package:base':
##
## transform
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v lubridate  1.9.4      v tibble    3.3.0
## v purrr      1.0.4      v tidyr     1.3.1
```

```
## -- Conflicts ----- tidyverse_conflicts() --
## x microbiome::alpha() masks ggplot2::alpha()
## x dplyr::filter()      masks stats::filter()
## x dplyr::lag()          masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggpubr)
library(knitr)
library(kableExtra)
```

```
##
## Attaching package: 'kableExtra'
##
## The following object is masked from 'package:dplyr':
##
##     group_rows
```

```
library(stringr)
library(patchwork)
library(viridis)
```

```
## Loading required package: viridisLite
```

```
source("/Users/mattparker/Dropbox/Workspace/moops/manuscript1/helper_utils.r")
knitr::opts_chunk$set(dev = "cairo_pdf")
```

Sequencing Summary Metrics

```
seq_metrics = read.csv("/Users/mattparker/Dropbox/Workspace/moops/manuscript1/data/sequencing_summary_m

seq_metrics %>%
  group_by(water_control) %>%
  summarize(Mean_reads = mean(passed_filters, na.rm = TRUE))
```

```
## # A tibble: 2 x 2
##   water_control Mean_reads
##   <chr>          <dbl>
## 1 No           3816698.
## 2 Yes          138170
```

Sample Taxon Reports

```
# List all CSV files in the directory
csv_files <- list.files(path = "/Users/mattparker/Dropbox/Workspace/moops/manuscript1/data/experiment_a
seq_metrics = read.csv("/Users/mattparker/Dropbox/Workspace/moops/manuscript1/data/sequencing_summary_m

# Process all CSV files and combine them
all_data <- bind_rows(lapply(csv_files, process_csv_file))

# Pond sample names
pond_samples = c("s9", "s10", "s19", "s20")

# Create sample metadata using helper function
sample_metadata = create_sample_metadata(seq_metrics, pond_samples)

# Merge the taxonomy data with metadata
merged_data <- all_data %>%
  left_join(sample_metadata, by = "sample_name")

# Filter to include only species level (tax_level == 1) to avoid duplication
species_data <- merged_data %>%
```

```

filter(tax_level == 1) %>%
filter(water_control == "No") %>% # filter out Negative Controls
filter(infection_class != "Definite") # filter out positive controls

species_data_filtered = species_data %>%
  filter(nt_rpm >= 1 | nr_rpm >= 1 | nt_count >=3) %>% # positive inclusion criteria
  filter(nt_alignment_length > 50)

# Experiment 1
seq_metrics_exp1 = sample_metadata %>%
  #left_join(sample_metadata, by = "sample_name") %>%
  filter(water_control == "No") %>% # Filter out Negative Controls
  filter(infection_class != "Definite") %>% # Filter out positive controls
  filter(!base_sample %in% c("s5-2")) %>%
  filter(experiment == 1) %>%
  mutate(
    total_reads_sub = total_reads * subsampled_fraction # Subsample total reads according to subsample
  )

seq_metrics_summary_1 = seq_metrics_exp1 %>%
  drop_na(base_sample) %>%
  select(base_sample, library_prep, total_reads, passed_filters, compression_ratio) %>%
  pivot_wider(
    names_from = library_prep,
    values_from = c(total_reads, passed_filters, compression_ratio),
    names_glue = "{.value}_{library_prep}"
  )

## Warning: Values from `compression_ratio`, `passed_filters` and `total_reads` are not
## uniquely identified; output will contain list-cols.
## * Use `values_fn = list` to suppress this warning.
## * Use `values_fn = {summary_fun}` to summarise duplicates.
## * Use the following dplyr code to identify duplicates.
## {data} |>
## dplyr::summarise(n = dplyr::n(), .by = c(base_sample, library_prep)) |>
## dplyr::filter(n > 1L)

# Experiment 3
seq_metrics_exp3 = sample_metadata %>%
  #left_join(sample_metadata, by = "sample_name") %>%
  filter(water_control == "No") %>% # Filter out Negative Controls
  filter(infection_class != "Definite") %>% # Filter out positive controls
  filter(!base_sample %in% c("s5-2")) %>%
  filter(experiment == 3) %>%
  mutate(
    total_reads_sub = total_reads * subsampled_fraction # Subsample total reads according to subsample
  )

seq_metrics_summary_2 = seq_metrics_exp3 %>%
  drop_na(base_sample) %>%
  select(base_sample, library_prep, total_reads, passed_filters, compression_ratio) %>%
  pivot_wider(

```

```

names_from = library_prep,
values_from = c(total_reads, passed_filters, compression_ratio),
names_glue = "{.value}_{library_prep}"
)

```

```
cat("\\captionsetup{labelformat=empty}")
```

```

# -----
# Calculate viral percentages per sample
# -----
sample_totals <- species_data %>%
  group_by(base_sample, library_prep) %>%
  summarise(
    total_nt_count = sum(nt_count, na.rm = TRUE),
    total_nr_count = sum(nr_count, na.rm = TRUE),
    total_nt_rpm   = sum(nt_rpm, na.rm = TRUE),
    total_nr_rpm   = sum(nr_rpm, na.rm = TRUE),
    .groups = "drop"
  )

virus_data <- species_data %>%
  filter(category == "viruses", is_phage == "false")

viral_percentages <- virus_data %>%
  drop_na(base_sample) %>%
  group_by(base_sample, library_prep) %>%
  summarise(
    viral_nt_count = sum(nt_count, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  left_join(sample_totals, by = c("base_sample", "library_prep")) %>%
  mutate(
    viral_nt_percent = (viral_nt_count / total_nt_count) * 100,
    viral_nt_percent = ifelse(is.nan(viral_nt_percent) | is.infinite(viral_nt_percent), NA, viral_nt_percent)
  )

viral_summary <- viral_percentages %>%
  select(base_sample, library_prep, viral_nt_count, viral_nt_percent) %>%
  pivot_wider(
    names_from = library_prep,
    values_from = c(viral_nt_count, viral_nt_percent),
    names_glue = "{.value}_{library_prep}"
  )

# -----
# Create seq_metrics_summary_1
# -----
seq_metrics_summary_1 <- seq_metrics_exp1 %>%
  drop_na(base_sample) %>%
  select(base_sample, library_prep, total_reads, passed_filters, compression_ratio) %>%
  group_by(base_sample, library_prep) %>% # ensure single row per sample+prep
  summarise(
    total_reads = sum(total_reads),

```

```

passed_filters = sum(passed_filters),
compression_ratio = mean(compression_ratio, na.rm = TRUE),
.groups = "drop"
) %>%
pivot_wider(
  names_from = library_prep,
  values_from = c(total_reads, passed_filters, compression_ratio),
  names_glue = "{.value}_{library_prep}"
) %>%
left_join(viral_summary, by = "base_sample") %>%
mutate(across(where(is.list), ~ as.numeric(unlist(.)))) %>% # convert any list-columns to numeric
mutate(
  passed_filters_msspe = (passed_filters_MSSPE / total_reads_MSSPE) * 100,
  passed_filters_ngs = (passed_filters_NGS / total_reads_NGS) * 100
) %>%
select(base_sample, total_reads_NGS, total_reads_MSSPE,
  passed_filters_ngs, passed_filters_msspe,
  compression_ratio_NGS, compression_ratio_MSSPE,
  viral_nt_percent_NGS, viral_nt_percent_MSSPE) %>%
arrange(desc(viral_nt_percent_MSSPE))

# Table 1: Experiment 1
seq_metrics_summary_1_kbl <- seq_metrics_summary_1 %>%
  rename(
    "Sample" = base_sample,
    "Reads (MSSPE)" = total_reads_MSSPE,
    "Reads (NGS)" = total_reads_NGS,
    "Passed Filter % (MSSPE)" = passed_filters_msspe,
    "Passed Filter % (NGS)" = passed_filters_ngs,
    "DCR (NGS)" = compression_ratio_NGS,
    "DCR (MSSPE)" = compression_ratio_MSSPE,
    "Viral Reads % (MSSPE)" = viral_nt_percent_MSSPE,
    "Viral Reads % (NGS)" = viral_nt_percent_NGS
  ) %>%
  kable(
    format = "latex",
    booktabs = TRUE,
    caption = "Experiment 1: Shallow-sequencing",
    digits = 2,
    align = "lrrr", # adjust as needed: text left, numbers right
    format.args = list(big.mark = ",")
  ) %>%
  kable_styling(
    latex_options = c("striped", "hold_position"), # polished journal look
    position = "center",
    full_width = TRUE,
    font_size = 9
  ) %>%
  row_spec(0, bold = TRUE) %>% # strong header
  collapse_rows(columns = 1, latex_hline = "major") # if repeating target names

# -----
# Create seq_metrics_summary_2 (Experiment 3)

```

```

# -----
seq_metrics_summary_2 <- seq_metrics_exp3 %>%
  drop_na(base_sample) %>%
  select(base_sample, library_prep, total_reads, passed_filters, compression_ratio) %>%
  group_by(base_sample, library_prep) %>%
  summarise(
    total_reads = sum(total_reads),
    passed_filters = sum(passed_filters),
    compression_ratio = mean(compression_ratio, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  pivot_wider(
    names_from = library_prep,
    values_from = c(total_reads, passed_filters, compression_ratio),
    names_glue = "{.value}_{library_prep}"
  ) %>%
  left_join(viral_summary, by = "base_sample") %>%
  mutate(across(where(is.list), ~ as.numeric(unlist(.)))) %>%
  mutate(
    passed_filters_msspe = (passed_filters_MSSPE / total_reads_MSSPE) * 100,
    passed_filters_ngs = (passed_filters_NGS / total_reads_NGS) * 100
  ) %>%
  select(base_sample, total_reads_NGS, total_reads_MSSPE,
    passed_filters_ngs, passed_filters_msspe,
    compression_ratio_NGS, compression_ratio_MSSPE,
    viral_nt_percent_NGS, viral_nt_percent_MSSPE) %>%
  arrange(desc(viral_nt_percent_MSSPE))

# Table 2: Experiment 3
seq_metrics_summary_2_kbl <- seq_metrics_summary_2 %>%
  rename(
    "Sample" = base_sample,
    "Reads (MSSPE)" = total_reads_MSSPE,
    "Reads (NGS)" = total_reads_NGS,
    "Passed Filter % (MSSPE)" = passed_filters_msspe,
    "Passed Filter % (NGS)" = passed_filters_ngs,
    "DCR (NGS)" = compression_ratio_NGS,
    "DCR (MSSPE)" = compression_ratio_MSSPE,
    "Viral Reads % (MSSPE)" = viral_nt_percent_MSSPE,
    "Viral Reads % (NGS)" = viral_nt_percent_NGS
  ) %>%
  kable(
    format = "latex",
    booktabs = TRUE,
    caption = "Experiment 2: Deep-sequencing",
    digits = 2,
    align = "lrrr", # adjust as needed: text left, numbers right
    format.args = list(big.mark = ",")
  ) %>%
  kable_styling(
    latex_options = c("striped", "hold_position"), # polished journal look
    position = "center",
    full_width = TRUE,

```

```

    font_size = 9
  ) %>%
  row_spec(0, bold = TRUE) %>% # strong header
  collapse_rows(columns = 1, latex_hline = "major") # if repeating target names

# Sequencing metrics ALL experiments
seq_metrics_all = sample_metadata %>%
  filter(water_control == "No") %>% # Filter out Negative Controls
  filter(infection_class != "Definite") %>% # Filter out positive controls
  filter(!base_sample %in% c("s5-2")) %>%
  mutate(
    total_reads_sub = total_reads * subsampled_fraction # Subsample total reads according to subsample
  )

seq_metrics_summary = seq_metrics_all %>%
  drop_na(base_sample) %>%
  select(base_sample, library_prep, total_reads, passed_filters) %>%
  pivot_wider(
    names_from = library_prep,
    values_from = c(total_reads, passed_filters),
    names_glue = "{.value}_{library_prep}"
  )

```

```

## Warning: Values from `passed_filters` and `total_reads` are not uniquely identified;
## output will contain list-cols.
## * Use `values_fn = list` to suppress this warning.
## * Use `values_fn = {summary_fun}` to summarise duplicates.
## * Use the following dplyr code to identify duplicates.
## {data} |>
## dplyr::summarise(n = dplyr::n(), .by = c(base_sample, library_prep)) |>
## dplyr::filter(n > 1L)

```

```

seq_metrics_all %>%
  group_by(library_prep) %>%
  summarize(mean_reads = mean(total_reads),
            mean_reads_sub = mean(total_reads_sub))

```

```

## # A tibble: 2 x 3
##   library_prep mean_reads mean_reads_sub
##   <chr>         <dbl>         <dbl>
## 1 MSSPE         41913532         5271475.
## 2 NGS           27105333         3616129.

```

```

seq_metrics_all_long <- seq_metrics_all %>%
  pivot_longer(
    cols = c(total_reads, total_reads_sub, passed_filters),
    names_to = "metric",
    values_to = "value"
  ) %>%
  mutate(metric = factor(metric, levels = c("total_reads", "total_reads_sub", "passed_filters")),
         value = value / 1e6, # Convert to millions of reads
         metric = recode(

```

```

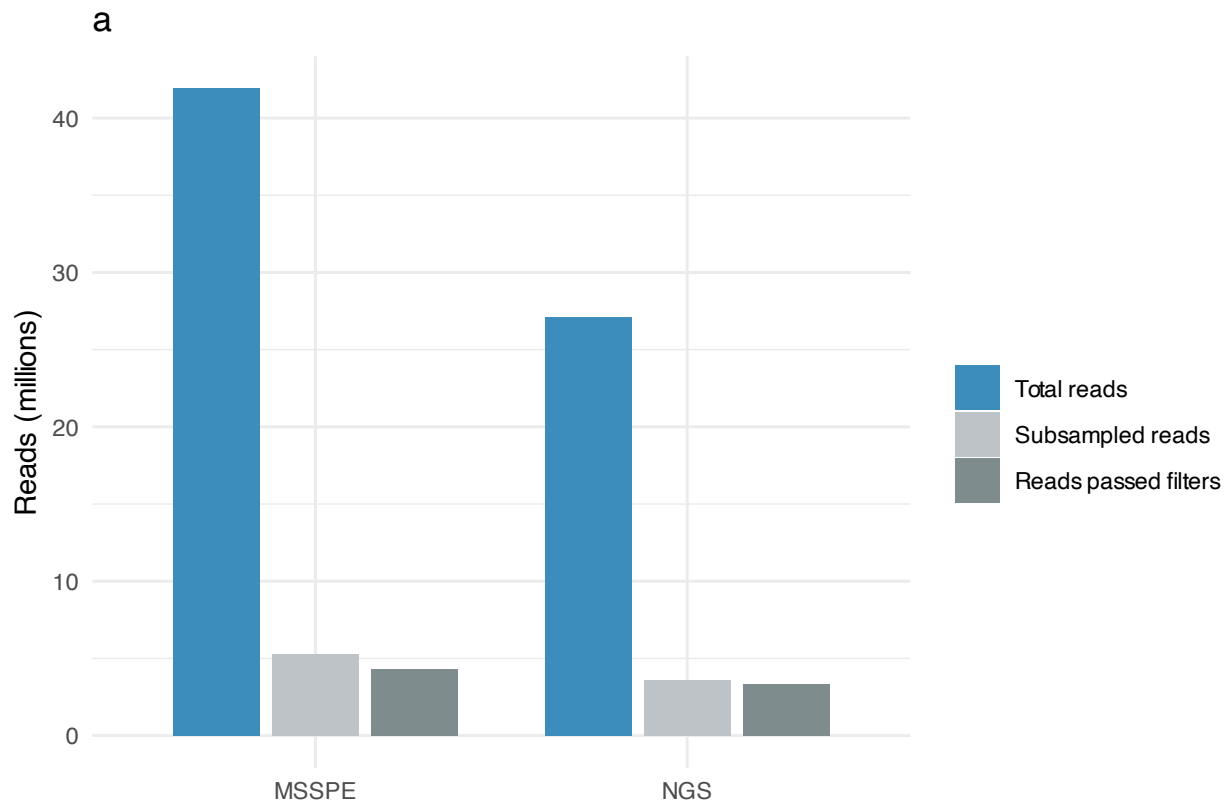
    metric,
    "total_reads" = "Total reads",
    "total_reads_sub" = "Subsampled reads",
    "passed_filters" = "Reads passed filters"
  )
)

plot_data <- seq_metrics_all_long %>%
  group_by(library_prep, metric) %>%
  summarize(mean_value = mean(value), .groups = "drop")

overall_barplot = ggplot(plot_data, aes(x = library_prep, y = mean_value, fill = metric)) +
  geom_bar(stat = "identity", position = position_dodge(width = 0.8), width = 0.7, linewidth = 0.2) +
  scale_fill_manual(
    values = c(
      "Total reads" = "#3C8DBC",    # professional blue
      "Reads passed filters" = "#7F8C8D", # neutral gray
      "Subsampled reads" = "#BDC3C7"    # light gray for secondary metric
    ),
    name = ""
  ) +
  labs(
    title = "a",
    x = "",
    y = "Reads (millions)"
  ) +
  theme_minimal()

overall_barplot

```

```
library(ggnewscale)
library(RColorBrewer)

# Create boxplot for viral NT percentage by control status
mean_df_viral_perc <- viral_percentages %>%
  group_by(library_prep) %>%
  summarise(mean_value = mean(viral_nt_percent, na.rm = TRUE))

viral_summary_wilcox = viral_summary %>%
  mutate(
    viral_nt_percent_NGS = replace_na(viral_nt_percent_NGS, 0), # set NAs to 0 for non detections
    viral_nt_percent_MSSPE = replace_na(viral_nt_percent_MSSPE, 0), # set NAs to 0 for non detections
  )

# Paired Wilcoxon test across all samples
wilcox_viral_perc <- wilcox.test(
  viral_summary_wilcox$viral_nt_percent_NGS,
  viral_summary_wilcox$viral_nt_percent_MSSPE,
  paired = TRUE,
  alternative = "two.sided"
)

pval_viral_perc <- wilcox_viral_perc$p.value

# get stupid color schemes so I cant make it continuous
blues_palette <- brewer.pal(9, "Blues") # Get a blues palette with 9 colors
reds_palette <- brewer.pal(9, "Reds") # Get a reds palette with 9 colors
```

```

library_prep_levels <- levels(factor(viral_percentages$library_prep))

viral_nt_boxplot <- ggplot(viral_percentages, aes(x = library_prep, y = viral_nt_percent)) +
  geom_boxplot(outlier.fill = "white", outlier.alpha = 0.5) +
  geom_jitter(aes(color = library_prep), width = 0.1, alpha = 0.7, size = 1.8) +
  scale_color_manual(
    values = c("MSSPE" = "#1f77b4", # blue
              "NGS" = "#d62728") # red
  ) +
  annotate("text", x = mean(1:length(library_prep_levels)), y = max(viral_percentages$viral_nt_percent,
    label = paste0("Wilcoxon p = ", signif(pval_viral_perc, 3)),
    fontface = "italic", size = 4) +
  stat_summary(fun = median, geom = "text", aes(label = round(..y.., 1)),
    vjust = -0.5, color = "black", fontface = "bold", size = 3) +
  geom_segment(data = mean_df_viral_perc,
    aes(x = as.numeric(library_prep) - 0.3,
        xend = as.numeric(library_prep) + 0.3,
        y = mean_value, yend = mean_value),
    linetype = "dotted", color = "black") +
  geom_text(data = mean_df_viral_perc,
    aes(x = library_prep, y = mean_value, label = paste0(": ", round(mean_value, 1))),
    vjust = -1, color = "black", size = 3.2, fontface = "italic") +
  theme_bw() +
  labs(
    title = "b",
    x = "",
    y = "% Viral Reads"
  ) +
  theme(
    #axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "none"
  )

viral_nt_boxplot

```

```

## Warning: The dot-dot notation (`..y..`) was deprecated in ggplot2 3.4.0.
## i Please use `after_stat(y)` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

```

```

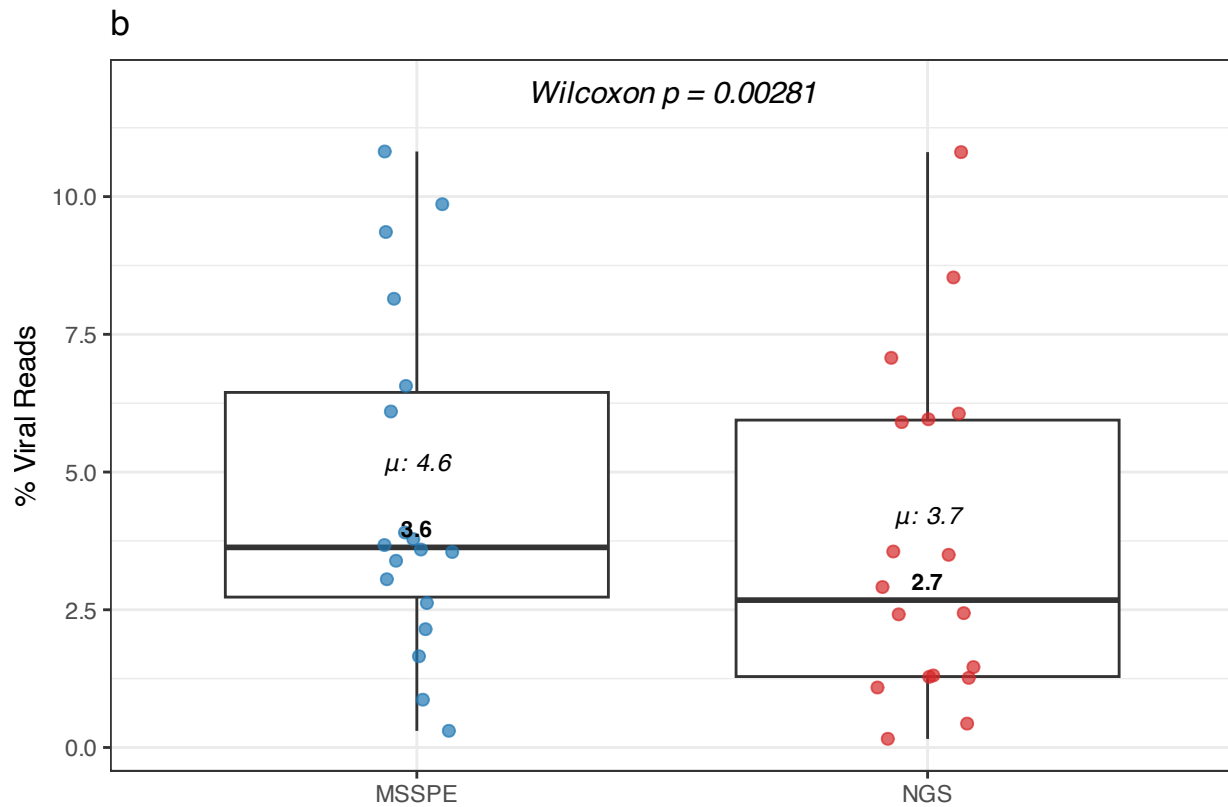
## Warning in FUN(X[[i]], ...): NAs introduced by coercion
## Warning in FUN(X[[i]], ...): NAs introduced by coercion

```

```

## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_segment()`).

```



```
horizontal_space <- plot_spacer() + plot_layout(widths = c(0.2))
```

```
combined_boxplots = (overall_barplot | horizontal_space | viral_nt_boxplot) +  
  plot_layout(ncol = 3, widths = c(1.5, 0.5, 1.5))
```

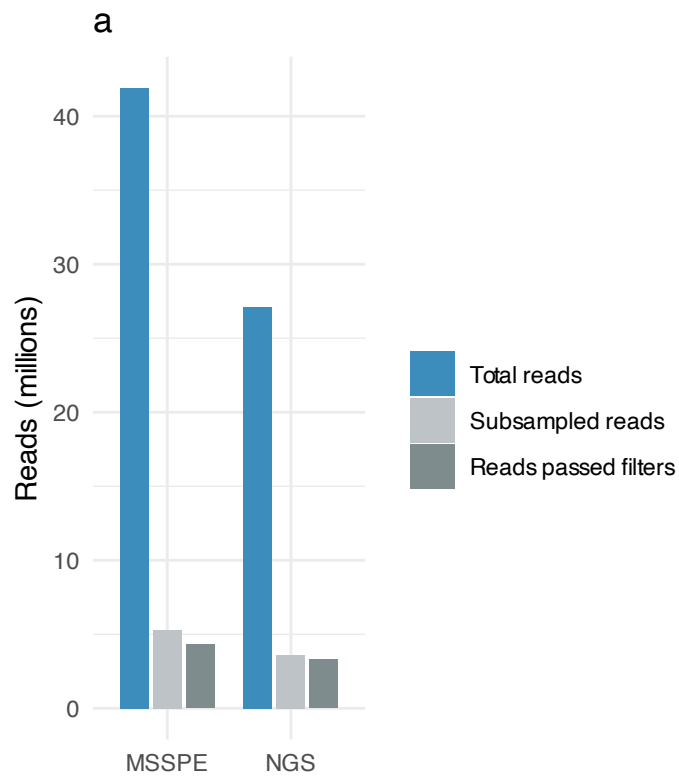
```
combined_boxplots
```

```
## Warning in FUN(X[[i]], ...): NAs introduced by coercion
```

```
## Warning in FUN(X[[i]], ...): NAs introduced by coercion
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
```

```
## (`geom_segment()`).
```



EXPORT ALL FIGURES AND TABLES FTW

